

# Finance Pipeline Database

## 3 Real-World Data Storage Examples

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# Overview: 3 Database Examples

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## Example 1

### AI-Extracted Market Events

NewsArticle → MarketEvent → Trend → Theme

How AI extracts 5 events from 1 article and cascades through the system

## Example 2

### Client Portfolio Holdings

Client → Account → Portfolio → Holdings

How we store \$500K portfolio with 5 securities

## Example 3

### Client Theme Matching

Portfolio Exposure → Market Themes

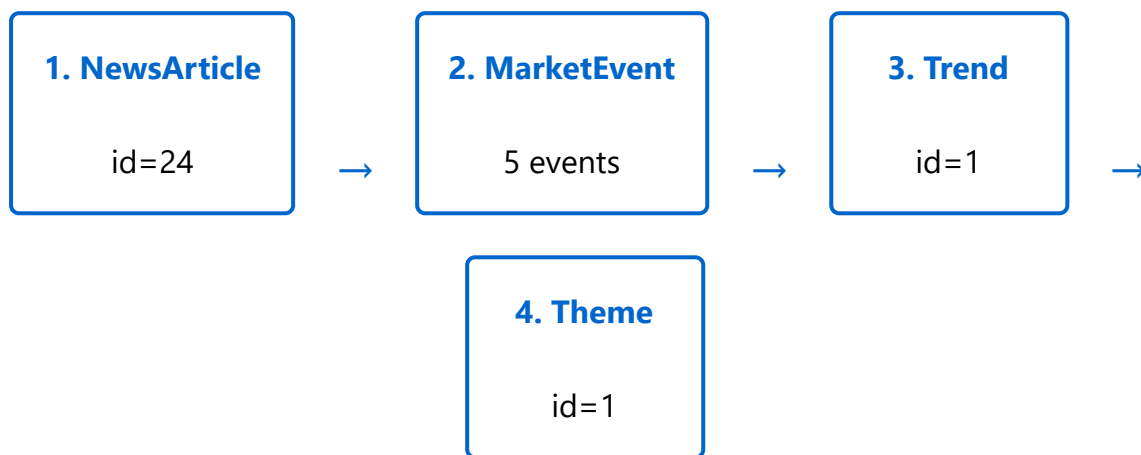
How we connect 62.52% portfolio exposure to market themes

# Example 1

AI-Extracted Market Events  
Linked to News Articles

# Data Flow: Article → Market Intelligence

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✓ **Result:** 1 Article → 5 Events → 1 Trend → 1 Theme → Tagged as POSITIVE for Technology

# NewsArticle & MarketEvent Tables

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## NewsArticle (id=24):

- Title: "Nvidia Just Slipped Below \$5 Trillion..."
- Processed: Yes (is\_processed=1)
- Ingested: 2026-06-08 04:53:35

## MarketEvent (5 discrete events from 1 article):

- Event 2: "Alphabet's stock pulled back... fell to \$4.45T" [Type: OTHER]
- Event 3: "Google Cloud revenue grew 63% YoY to \$20B" [Type: EARNINGS]
- Event 4: "Order backlog nearly doubled to \$460B" [Type: OTHER]
- Event 5: "Google search & ad revenue grew 19%" [Type: EARNINGS]

**Entities stored as JSON:** ["Alphabet", "GOOG"]

# Trend → Theme → SectorTag

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## Trend (id=1):

- Name: "Alphabet Growth and Stock Adjustments"
- Direction: **BULLISH** ↑

## Theme (id=1):

- Name: "Tech Resurgence"
- Description: Technology sector witnessing revival driven by cloud & advertising growth

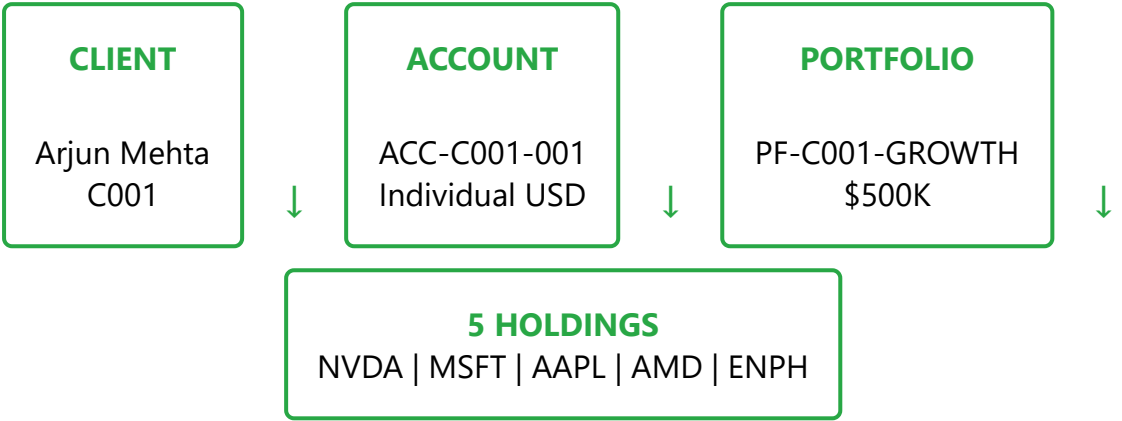
## SectorTag (Theme linked to Sector + Sentiment):

- Theme: Tech Resurgence
- Sector: **Technology**
- Sentiment: **POSITIVE** ✓
- Confidence: **0.9 (90%)**

# Example 2

Client Portfolio Holdings  
with Security Prices

# Portfolio Hierarchy



# Client & Account Information

### Client Table (id=1)

- Code: **C001**
- Name: **Arjun Mehta**
- Email: arjun.mehta@example.com
- Risk Profile: **Aggressive**
- Created: 2026-06-07

### Account Table (id=1)

- Number: **ACC-C001-001**
- Type: **INDIVIDUAL**
- Currency: **USD**
- Active: **Yes**
- Opened: 2026-06-07

**Portfolio:** PF-C001-GROWTH | Strategy: Growth | Value: **\$500,000** | Inception: 2023-01-01

# Portfolio Holdings Breakdown (\$500K)

| Ticker | Qty | Price | Value     | Weight  | Sector           |
|--------|-----|-------|-----------|---------|------------------|
| NVDA   | 200 | \$900 | \$180,000 | 36.0% ★ | Semiconductors   |
| MSFT   | 150 | \$430 | \$64,500  | 12.9%   | Technology       |
| AAPL   | 100 | \$195 | \$19,500  | 3.9%    | Technology       |
| AMD    | 300 | \$162 | \$48,600  | 9.72%   | Semiconductors   |
| ENPH   | 50  | \$115 | \$5,750   | 1.15%   | Renewable Energy |



**Sector Allocation:** Tech/Semiconductors 72% | Renewable 9.7% | Other 18.3%

# Example 3

Client Theme Matching  
Portfolio to Market Intelligence

# Pipeline: News → Client Exposure

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## MARKET SIDE

**Theme:** "Tech Resurgence"

**SectorTag:**

- Sector: Technology
- Sentiment: **POSITIVE**
- Confidence: **0.9 (90%)**

## CLIENT SIDE

**Arjun's Portfolio:**

- 200 NVDA = \$180K
- 150 MSFT = \$64.5K
- 100 AAPL = \$19.5K
- 300 AMD = \$48.6K

**Total Tech:** \$312.6K (62.52%)

**CONNECTION:** Market Theme directly connects to portfolio exposure!

# ClientThemeMatch Analysis

**Client:** Arjun Mehta (C001) | **Theme:** "Tech Resurgence" (ID: 1)

✔ **Matched Sectors from Portfolio:** Technology, Semiconductors

💰 **Portfolio Exposure Calculation:**

- NVDA:  $200 \times \$900 = \$180,000$
  - MSFT:  $150 \times \$430 = \$64,500$
  - AAPL:  $100 \times \$195 = \$19,500$
  - AMD:  $300 \times \$162 = \$48,600$
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- **TOTAL: \$312,600 (62.52% of \$500K portfolio)**

📊 **Exposure Metrics:**

- Sentiment: **POSITIVE** ✔
- Confidence: **0.9 (90%)**

🎯 **INSIGHT:** 62.52% of Arjun's \$500K portfolio is directly exposed to POSITIVE 'Tech Resurgence' theme with 90% confidence!

# Complete Data Flow Architecture

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**LAYER 1 → INPUT:** NewsArticle - Raw articles from RSS feeds

**LAYER 2 → AI EXTRACTION:** MarketEvent - Discrete events via AI (1 article = 5 events)

**LAYER 3 → PATTERN:** Trend - Cross-article patterns (e.g., "Alphabet Growth")

**LAYER 4 → NARRATIVE:** Theme - Investment narratives (e.g., "Tech Resurgence")

**LAYER 5 → TAGGING:** SectorTag - Sector + Sentiment + Confidence

**LAYER 6 → MATCHING:** ClientThemeMatch - Portfolio exposure analysis

**LAYER 7 → PORTFOLIO:** Holding - Actual client positions



**RESULT: Market Intelligence → Client Portfolios → Actionable Insights**

# Key Insights & Takeaways

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- ✓ **Multi-Layer Processing:** Each article yields 5+ events cascading through trends, themes, and sector tags
- ✓ **Hierarchical Organization:** Client → Account → Portfolio → Holdings structure enables portfolio-level analytics
- ✓ **Real-Time Portfolio Insights:** Market themes connect directly to client holdings for personalized recommendations
- ✓ **Confidence Scoring:** All AI insights include confidence scores (0.0-1.0) for reliable decision-making
- ✓ **Scalable Architecture:** Database design supports unlimited clients, portfolios, themes, and market events

# Thank You

Finance Pipeline Database Examples

Connecting Market Intelligence to Client Portfolios